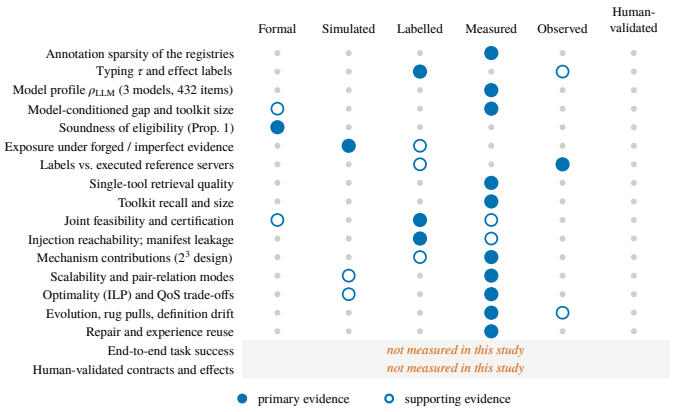


Figure 8. Evidence map: the class of evidence behind each claim evaluated in this section (filled: primary, open: supporting). Shaded rows are not measured.

Table 7. Data, splits and their use.

Data	Size	Use
Merged registry	569 / 76	tools / servers; 44 annotated; all experiments
LiveMCPBench tasks	95	multi-step test tasks (94 with gold tool)
MCPToolBench++ dev	60	tuning of every method (10 per category)
MCPToolBench++ test	300	single-tool retrieval (50 per category, 66 targets)
Security task set	155	95 test tasks + 60 dev prompts, two policies
Calibration suite	432	6 operations × 4 sizes × 2 languages × noise × context; 264 calibration / 168 test items, 3 models
Reference servers	6 / 38	servers / registry tools executed under tracing
Cloned registries	10^3–10^5	near-copies of real contracts (scaling)
Heterogeneous registries	10^3–10^5	generator varying servers, effects, sparsity, vocabulary, duplicates
Cover instances	80	8 families × 10; exact optima by ILP
QoS instances	120	4 roles × 4 providers; 4 weightings × 3 floors

8 Experimental Evaluation

This section evaluates PaMCP along the four research questions of Section 1. Section 8.1 fixes the data, the compared methods, the varied parameters and the statistical protocol. The next five subsections answer RQ1 (representation and model-conditioned gap, Section 8.2), RQ2 (trust under missing, forged and imperfect evidence, Section 8.3), RQ3 (retrieval, toolkit quality and certification, Section **??**), the security behaviour and the contribution of each mechanism (Section **??**), and cost, scalability, adaptation and stability (RQ3/RQ4, Section **??**). Section **??** states the limits of the evidence. Every subsection gives the protocol, a results table, figures, the observations and their interpretation, and compares PaMCP with baselines wherever a baseline exists.

8.1 Experimental Setup

Evidence classes. The claims of the paper differ in what can establish them. We label every result with one of six evidence classes: *formal* (a proof in Sections 5–7), *simulated* (generated under stated assumptions, e.g. a verifier with a controlled error rate), *labelled* (scored against machine-generated reference labels), *measured* (obtained from the running implementation against benchmark gold or solver optima), *observed* (from controlled execution traces of real MCP servers) and *human-validated*. Fig. 8 maps each claim to the classes that support it. Two claims have no supporting evidence in this study: end-to-end task success of executed toolkits and human-validated contracts; no conclusion below rests on them.

Registries and tasks. Table 7 lists the data. The registry merges LiveMCPTool [1] and the tool set of MCP-ToolBench++ [2] by `server::name` (76 servers, 569 tools; Fig. 9a). Only 44 tools carry MCP annotations, all of them `readOnlyHint=true`, and one of these is not read-only (Fig. 9c); 119 of the 525 unannotated tools are not read-only. Multi-step tasks are the 95 public LiveMCPBench tasks (94 with a resolvable gold tool). Single-tool retrieval uses MCP-ToolBench++ requests: a 60-request development split (ten per category) for every tuning decision, and a disjoint test sample of 300 requests (50 per category, 66 target tools). Security and ablation experiments use 155 tasks (the 95 LiveMCPBench tasks and the 60 development prompts), each under its least-privilege policy and under a strict read-only policy; confirmatory tests use the test tasks only.

Reference labels. Neither benchmark labels effects or types. Labels J_A (Claude Sonnet, from name, description and schema under a fixed guide) give the typing τ and drive the simulated verifier bound to $h(t)$; labels J_B (Claude Opus, blind, separate guide) are used for scoring (Fig. 9b). They agree on state modification for 558 of 569 tools ($\kappa = 0.944$). Because every tool that is unsafe under J_B (120) is also unsafe under J_A (131), a verifier simulated from J_A can never admit a tool that J_B calls unsafe; we therefore also report the swapped direction (Section 8.3) and compare both labellings with executed servers. All LLM outputs (analyst goal contracts, MCP-Zero requests, COLT training queries, ToolScope merge judgements, calibration answers) were recorded once and are replayed.

Compared methods. Table 8 lists the methods and what each may use. PaMCP combines three mechanisms that the ablations switch independently: the model-conditioned gap (G, Algorithm 1), provenance-sound eligibility (P, Section 5) and set-level reasoning with certification (S: backward closure, minimum-cost cover, certificate; Section 6). PaMCP-L is the relevance-first variant selected on the development split (lexical candidate ranking; structural matching only as a recorded, uncertain constraint). The baselines are adapted to the common registry and share one dense encoder (TF-IDF with truncated SVD); they are adaptations, not the published systems. Every retrieval method was tuned on the development split with a comparable grid (Table 9).

Varied parameters. Table 9 lists every parameter that an experiment varies, with its range and default; parameters not listed keep the values of Section 6.

Statistical protocol. Request-level metrics are resampled with target tools as clusters and task-level metrics with tasks as clusters; paired comparisons use two-sided sign-flip permutation tests on cluster sums, proportions use Wilson intervals, and all intervals are 95%. Two confirmatory families were fixed before the data they test were seen: seven comparisons F1–F7 (before the set-level experiments) and an amendment H1–H5 (committed, dated, before any test-set run of the revised analy-

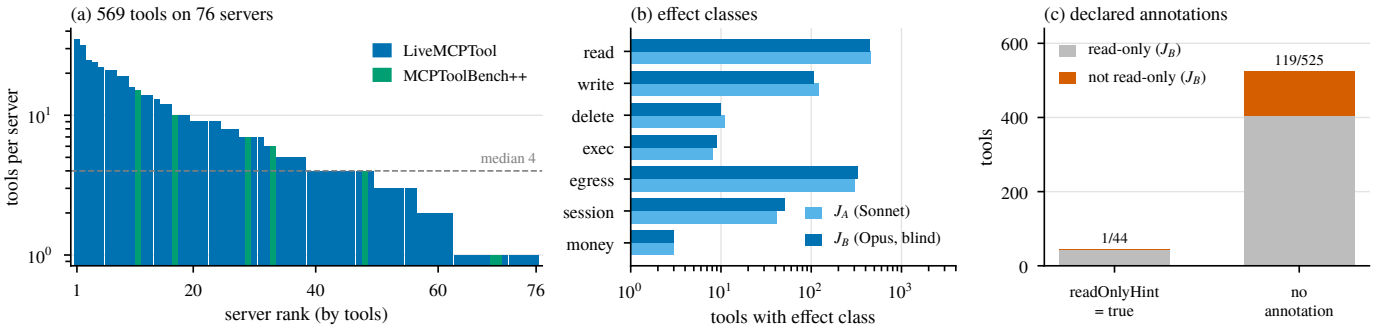


Figure 9. The evaluation registry. (a) Tools per server (log scale). (b) Effect classes under the two LLM labellings. (c) Declared annotations against the read-only status under J_B (numbers: not read-only / total).

Table 8. Compared methods: information used and tuning on the development split.

Method	Information used	Grid
PaMCP	analyst goal contracts, typing τ, evidence, profile, Eq. (6)	14
PaMCP-L	as PaMCP; lexical ranking first, types as constraint	(sel.)
RAG-MCP [3]	raw request, dense top-k	1
BM25 / RRF fusion	raw request, lexical / lexical+dense	12 / 4
MCP-Zero [4]	model-written requests, server routing	4
COLT [5]	query–scene–tool graph of 450 training queries	16
ToolScope [6]	LLM-confirmed tool merging, analyst subqueries	3
CMTF [7]	typed-state search, one tool per step	4
Per-subtask	analyst subgoals, one tool each	–
Full catalogue	all 569 tools	–

Table 9. Parameters varied in the experiments (default in bold).

Parameter	Values	Section
Candidate budget K	1, 5, **10**, 20, 40	**??**, **??**
Relevance weight w_{rel}	0, 1, 3, **10**	**??**
Profile threshold τ	0.6, 0.75, **0.9**; risk targets 0.05, 0.10	8.2
Input size n	10, 30, 40, 160, 300, 640, 3k, 30k	8.2
Forging servers δ	0, 0.05, 0.1, 0.2, 0.3 (5 seeds each)	8.3
Verifier false accept φ	0, 0.01, 0.02, **0.05**, 0.1, 0.2	8.3
Probe coverage κ	0.25, 0.5, 0.75, **1**	8.3
Trust threshold $\ell_{\min}$	0.2–0.95 (step 0.05), **0.65**	8.3
Poisoned descriptions	0, 10%, 30% of tools	**??**
Mechanisms G, P, S	on/off (2^3) + 5 components	**??**
Registry size	569, 10^3, 10^4, 10^5 tools	**??**
Pair relation	**eager**, partitioned, lazy	**??**
Cost weights $\beta, \gamma, \delta, \eta, \zeta$	0, 0.25, **0.5**, 1	**??**
Utility floor $u_{\min}$	0, 0.4, **0.5**, 0.6, 0.8, 0.9	**??**
Node budget (exact search)	**20,000**	**??**

ses), each with Holm correction and, for recall, a non-inferiority margin of 0.05. All other comparisons are exploratory and reported as such. Code, recorded LLM outputs, per-item results, run manifests (commit, input and label hashes, package versions) and a notebook that re-runs every experiment on a CPU runtime are available.[1]

8.2 RQ1: Representation and Model-Conditioned Gap

RQ1 asks whether the capability model can represent what tools and models do well enough to decide what a request needs. We

[1] `https://github.com/msellamiTN/taas-provenance-discovery`, branch `research/pamcp-final-evaluation`.

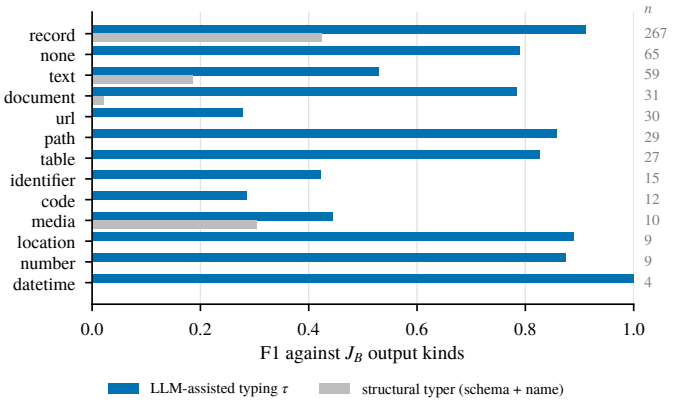


Figure 10. Typing agreement with J_B per output kind (kinds with at least four tools; right: support). The LLM-assisted typing τ is compared with a structural typer that uses schemas and names only.

test three of its ingredients: the typing τ on which structural reasoning relies, the measured reliability profile ρ_{LLM} that decides which operations the model keeps, and the gap of Algorithm 1 that turns the two into a toolkit.

Protocol. Typing is scored against the blind labels J_B (labelled evidence). The profile is measured on a calibration suite of 432 generated data operations (extraction, classification, counting, approximate sums, exact computation and joins; 10 to 640 records; English and French; clean or noisy text; isolated or embedded in a multi-step context). Claude Haiku, Sonnet and Opus answered every item by reading only, without tools, and stated beforehand, per cell, their probability of answering correctly. The threshold τ is chosen on 264 calibration items as the smallest value whose selective risk, the false-native rate among operations kept native, is at most the risk target, and is evaluated on 168 held-out items (measured evidence). One Opus batch of 45 items was interrupted by a usage limit; its items are excluded for Opus, not counted as errors. The gap is exercised on the review scenario of Fig. 1 for growing inputs.

Typing. The LLM-assisted typing agrees with J_B on 78.9% of output kinds ($\kappa = 0.718$, macro-F1 0.704) but on only 35.7% of complete types (kind and domain). A structural typer working from schemas and names alone reaches 22.8% ($\kappa = 0.070$, macro-F1 0.062) and recognises only records, text, documents and media at all (F1 at most 0.42; Fig. 10); τ itself is weakest

Table 10. Model-profile calibration: accuracy (Wilson 95% CI), self-assessment calibration, and the held-out behaviour of the threshold τ chosen for a 5% risk target (measured profile) versus the 64-item profile at $\tau = 0.9$. Cov.: share of operations kept native; FNR: false-native rate (errors among them).

				Self-assess.		Measured, risk 5%			64-item	
Model	n	Accuracy (%)	n=640	ECE	Brier	τ	Cov.	FNR	Cov.	FNR
Haiku	432	86.1 [82.5, 89.1]	67	0.39	0.30	0.21	83	3.6	21	0.0
Sonnet	432	91.4 [88.4, 93.7]	62	0.20	0.11	0.42	90	3.3	26	4.5
Opus	387	100.0 [99.0, 100.0]	100	0.38	0.20	0.90	100	0.0	–	–

on URLs, code and identifiers (F1 0.28–0.42). Types are therefore reliable at the kind level and unreliable at the domain level, which is where structural matching discriminates; this limits structural retrieval (Section **??**).

Capability boundaries. Table 10 and Fig. 11 show that the boundary between native and tool-requiring operations depends on the model and on the operation, not only on the input size. Haiku answers 86.1% of the items (95% CI 82.5–89.1%) and fails counting already at medium sizes (21% at $n = 40$, 6% at $n = 160$); Sonnet answers 91.4% (88.4–93.7%) and loses joins as they grow (71% at $n = 40$, 44% at $n = 160$); both fail approximate sums over 640 numbers. Opus answers all 387 items it was given, so its boundary lies beyond the suite. Language matters little (Haiku 88.0% in English against 84.3% in French; Sonnet 91.2% against 91.7%), whereas noise lowers Sonnet from 95.4% to 87.5% and exactness lowers Haiku from 92.5% (tolerant) to 78.1% (exact). A single profile per model family would thus misplace the boundary in both directions.

Measured against self-assessed reliability. All three models underestimate themselves (Fig. 12a): their stated probabilities average 0.47, 0.72 and 0.62 against accuracies of 0.86, 0.91 and 1.00 (ECE 0.39, 0.20 and 0.38). With the measured profile and a 5% risk target, the held-out false-native rate stays within the target for every model (3.6%, 3.3% and 0.0%) while 83%, 90% and 100% of the operations remain native (Fig. 12b). The 64-item profile used before this revision is safe but wasteful at $\tau = 0.9$: it keeps only 21% (Haiku) and 26% (Sonnet) of the operations native and sends most of the work the models perform correctly to tools. Self-assessment cannot replace measurement: at the same target it keeps 24% of Haiku's operations native. On the test and development tasks the measured profile overrules 12 of the analyst's 128 claims that an operation is native (9.4%).

Model-conditioned gap. On the review scenario the gap behaves as designed (Fig. 12c): with tolerant aggregation two tools suffice at $n = 30$ and 300, where the model keeps 60% of the goals, and six are needed at $n = 3{,}000$; exact aggregation adds two tools at the small sizes; at $n = 30{,}000$, and at $n = 3{,}000$ with $\tau = 0.6$, the engine abstains instead of returning an unsupported toolkit. Without gap analysis it abstains in every setting, because no registry tool serves the scenario's interpretive goals.

Interpretation. The representation supports model-conditioned discovery: what a request needs depends on the deployed model, and a measured profile with a risk-targeted threshold bounds the false-native rate on held-out operations at the chosen level. Two limits remain. The suite covers six data operations, not the whole transformation vocabulary of Definition 6, and Opus reaches its ceiling; profile stability across model updates was not measured. Typing is the weaker ingredient: its domain-level disagreement propagates to every structural decision evaluated next.

Table 11. Encodings of a read-only policy over 273 requests with read-only gold tools: eligible share of the registry, unsafe tools admitted (of 120, J_B), exposure and utility of the top five for PaMCP's retrieval stage and MCP-Zero, and exposure when 20% of the servers forge and launder their metadata (verifier false-accept rate 5%, mean of five seeds).

			PaMCP retr.		MCP-Zero		δ=0.2
Encoding	Elig.	Unsafe	Exp@5	Util@5	Exp@5	Util@5	Exp@5
no filter	100.0	120	42.5	49.8	26.4	76.6	43.7
MCP default (display only)	100.0	120	42.5	49.8	26.4	76.6	43.7
closed-world annotations	7.7	1	31.5	0.4	0.0	0.4	58.7
lexical cues	67.8	21	9.9	41.0	12.5	65.6	8.8
verified evidence (P)	77.0	0	0.0	46.2	0.0	72.2	5.5
observed + lexical fallback	77.0	0	0.0	46.2	0.0	72.2	5.5
attested only	26.4	0	0.0	16.5	0.0	18.3	5.8

8.3 RQ2: Trust under Missing, Forged and Imperfect Evidence

RQ2 asks whether provenance-sound eligibility (P) keeps unsafe tools away from the model when metadata is missing, forged or verified imperfectly, and what this costs in utility. Proposition 1 guarantees soundness only if safety-relevant evidence is truthful (formal); the experiments below measure how exposure behaves when that assumption is relaxed (simulated), whether the result depends on how the reference labels were built (labelled), and whether the labels agree with what real servers do when executed (observed).

Protocol. A read-only policy is applied to the 394 requests of the single-tool test sample and the development split; Exp@5 is the share of the 273 requests with read-only gold tools whose top five contain a tool that is not read-only under J_B, and Util@5 the share whose top five contain a gold tool. Each encoding of the policy is applied to PaMCP's retrieval stage and to MCP-Zero. Forgery is simulated by letting a share δ of the servers declare `readOnlyHint=true` on every tool and launder their descriptions (five seeds per δ); the verifier is simulated with false-accept rate φ, false-reject rate ψ and probe coverage κ. For the observed comparison, the six MCP reference servers of the registry were started from their published packages (current versions) in a sandbox; each of their 37 registry tools was called once with schema-derived arguments under system-call tracing and file-system snapshots, and a state change was recorded when the call wrote, renamed or deleted user files, the server's store or version-control content.

Declared metadata is not evidence. Table 11 compares the encodings at full verification coverage. Without a filter, and under MCP's default of displaying annotations without enforcing them, 42.5% of the requests receive an unsafe tool among PaMCP's top five (26.4% for MCP-Zero). Treating the annotations as a closed world keeps only the 44 annotated tools; the one unsafe tool among them then reaches 31.5% of the lists while utility collapses to 0.4%. Lexical cues admit 21 of the 120 unsafe tools (Exp@5 9.9%, Util@5 41.0%; they flag 82.5% of

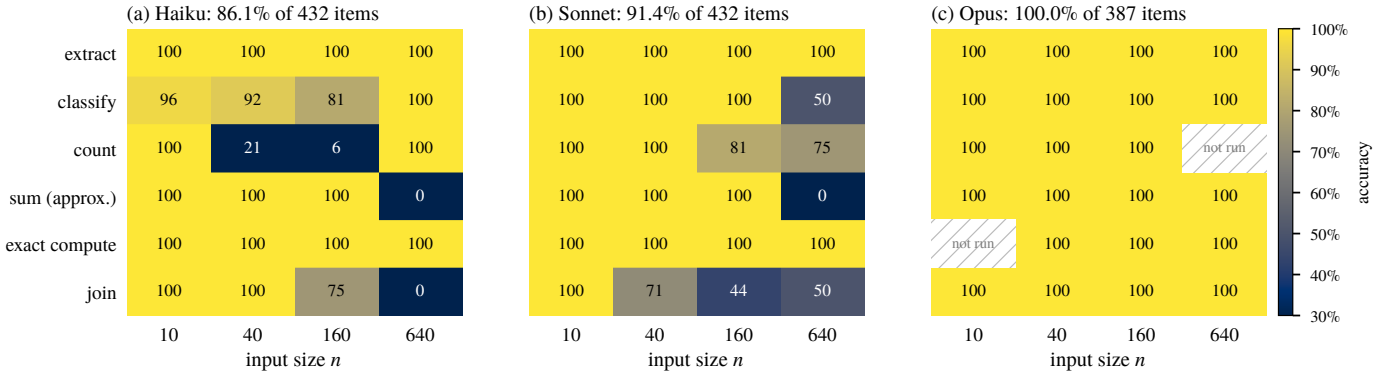


Figure 11. Measured reliability per operation and input size for three models (calibration suite, both splits; cells average languages, noise and context). Hatched cells belong to the interrupted Opus batch.

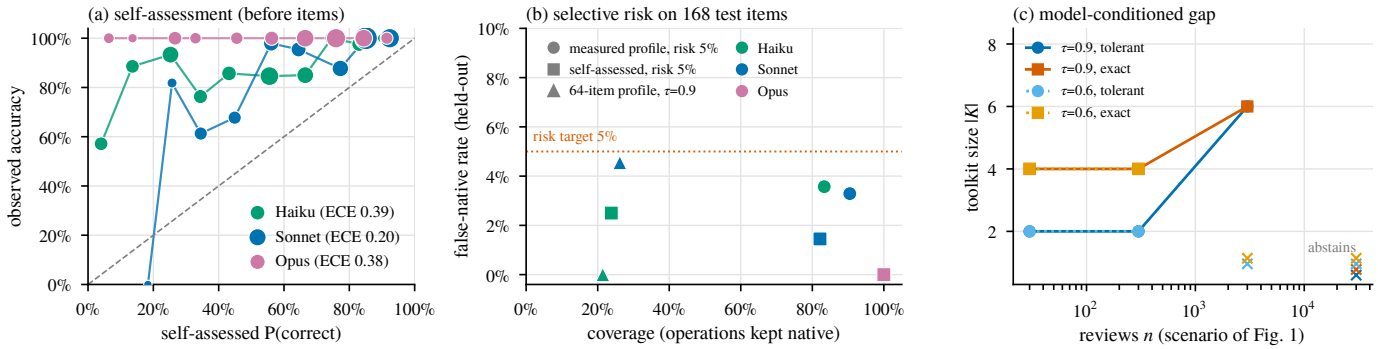


Figure 12. (a) Reliability diagram of the models' self-assessment, given before any item was seen (marker area: items per bin). (b) Held-out coverage and false-native rate of three profiles for each model. (c) Toolkit size of the review scenario against the input size, Sonnet profile; crosses mark abstentions.

the unsafe tools at a precision of 0.54). Verified evidence admits none, keeps 77.0% of the registry eligible and costs 3.6 points of utility for PaMCP's retrieval (46.2% against 49.8%) and 4.4 points for MCP-Zero (72.2% against 76.6%); with lower probe coverage, exposure stays at zero and utility follows the eligible share (Fig. 13a).

Forged and imperfect evidence. Forgery breaks the closed-world reading: its exposure rises from 31.5% to 57.1% when a tenth of the servers forge and to 62.6% at 30% (Fig. 13b). Laundered descriptions barely affect lexical cues (8.7–9.9%), because tool names carry most cues. Verified evidence is independent of δ by construction of the probe: with a 5% false-accept rate its exposure is 5.4–5.6% at every δ, including $\delta = 0$ (five seeds each), at a utility of 46.1–46.7%; which unsafe tools are falsely accepted varies with the seed (0.0–14.7% per seed). Exposure grows roughly linearly with the false-accept rate (1.1%, 5.7%, 9.3% and 15.8% at φ of 1, 5, 10 and 20%; Fig. 13c); errors concentrated on look-alike unsafe tools are not worse. Verified evidence stays below the lexical encoding (9.2%) up to $\varphi = 5\%$ with random errors and up to 10% with look-alike errors: a verifier must be at least this accurate to help. False rejections cost utility instead ($\psi = 0.2$: 46.2% to 39.1%).

Trust threshold. Raising $\ell_{\min}$ from 0.55 to 0.65, the level of observed evidence, lowers exposure from 5.5% to 3.5% and utility from 45.8% to 34.4%; requiring $\kappa \geq 0.8$ as well lowers them to 0.6% and 14.3% (Fig. 14a). Above 0.75 only attested evidence qualifies and exposure rises again to 10.3%, because the few falsely accepted attested tools reach many lists of a pool of 25.4% of the registry. A stricter threshold therefore does not lower exposure by itself; exposure must be read with the eligible pool.

Table 12. State changes observed when executing the 30 exercised tools of six MCP reference servers, against the two labellings and the servers' current annotations. FS: labelled read-only but observed to modify state (false safe; Wilson 95% CI); Unconf.: labelled state-changing, not triggered by the probe.

Reference	TP	FS	Unconf.	TN	False-safe rate (%)	Agree.
J_A (Sonnet)	13	0	0	17	0 [0, 22.8]	100
J_B (Opus)	13	0	0	17	0 [0, 22.8]	100
live annotations	13	0	5	12	0 [0, 22.8]	83

Independence of the reference labels. With the verifier simulated from J_A and scoring on J_B, zero exposure at $\varphi = 0$ is guaranteed by the inclusion of the unsafe sets and carries no empirical content. Swapping the roles removes the inclusion: a verifier simulated from J_B admits the 11 tools that only J_A considers unsafe, and verified evidence then exposes 2.3% of the requests for PaMCP's retrieval, RAG-MCP and ToolScope and 3.9% for MCP-Zero (Fig. 14b), against 9.9% for lexical cues under the original scoring. The protection therefore survives a disagreement between independent labellers, at a non-zero rate that measures their disagreement rather than the verifier.

Observed effects of executed servers. All six reference servers started; 30 of the 37 probed tools completed their call

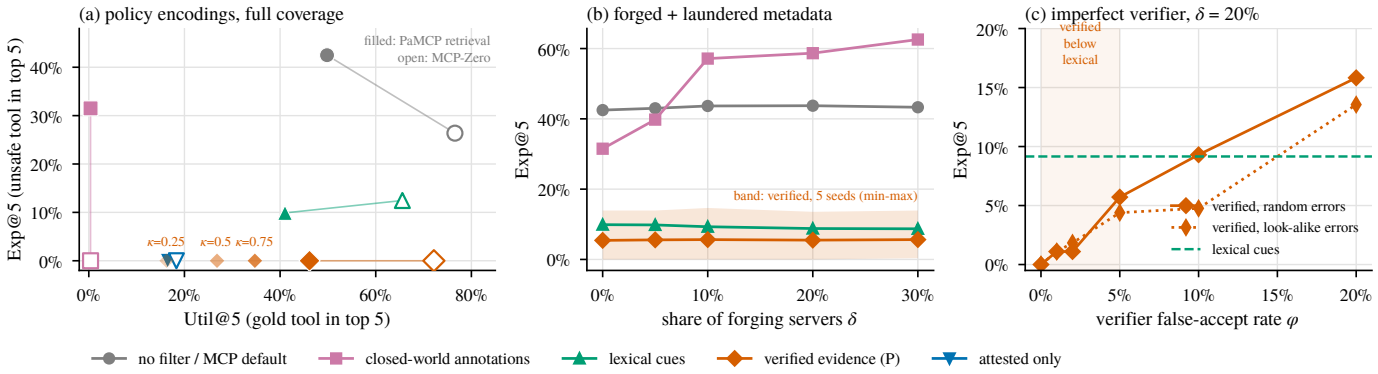


Figure 13. Exposure under a read-only policy (273 requests). (a) Exposure against utility per encoding at full coverage, and verified evidence at lower probe coverage κ. (b) Exposure as a share δ of servers forges annotations and launders descriptions (verifier false-accept rate 5%; mean of five seeds, band: range of the seeds for verified evidence). (c) Exposure of verified evidence against the false-accept rate, with errors at random or on look-alike unsafe tools, against the lexical encoding.

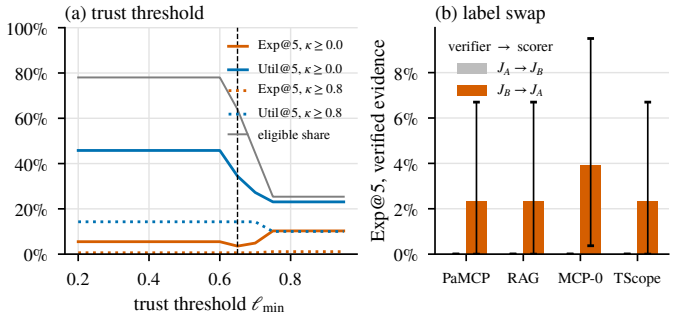


Figure 14. (a) Exposure, utility and eligible share against the trust threshold $\ell_{\min}$ ($\delta = 0.2$, $\varphi = 5\%$), with and without a minimum probe coverage $\kappa \geq 0.8$. (b) Exposure of verified evidence when the verifier is simulated from one labelling and scored on the other (cluster-bootstrap 95% CI).

(six file-system calls received arguments the synthesiser could not make valid, e.g. an edit whose target text is absent, and the fetch call was blocked by the sandbox's network policy), and 13 of them changed state: the memory server's six writers rewrite its store, five Git tools commit, stage, reset, branch or check out, and two file-system tools write or move files (Fig. 15a). Both labellings agree with every exercised call (Table 12): no tool observed to change state is labelled read-only (0 of 13; Wilson upper bound 22.8%), and no label of a state change went unconfirmed. The servers now publish annotations for 25 of these tools, none in the registry snapshot; none claims read-only for an observed state change, and they are conservative on 5 read-only tools. Since the snapshot, every one of the 37 surviving definitions has changed (annotations 25, input schemas 19, descriptions 8; Fig. 15b), one tool was removed and five were added; 13 file-system tools now publish input schemas without properties, which the official Python SDK rejects. Bound to $h(t)$, every piece of evidence about these servers would have been invalidated and re-established before use.

Interpretation. Provenance-sound eligibility removes the exposure that declared or lexical metadata leaves (9.9–42.5%) at a utility cost of 3.6–4.4 points, and degrades gracefully, roughly in proportion to the verifier's false-accept rate, when evidence is imperfect. The zero exposure of an error-free verifier is a property of the simulation; the swapped labels bound the effect of labeller disagreement at 2.3–3.9%, and execution confirms the labels on the state changes of 30 reference tools. These servers are maintained by the protocol's authors and are not adversarial; behaviour against malicious third-party servers is outside the observed evidence (Section **??**). The drift measurement supports binding evidence to $h(t)$: in the ecosystem studied, definitions change often enough that unbound evidence would be stale.

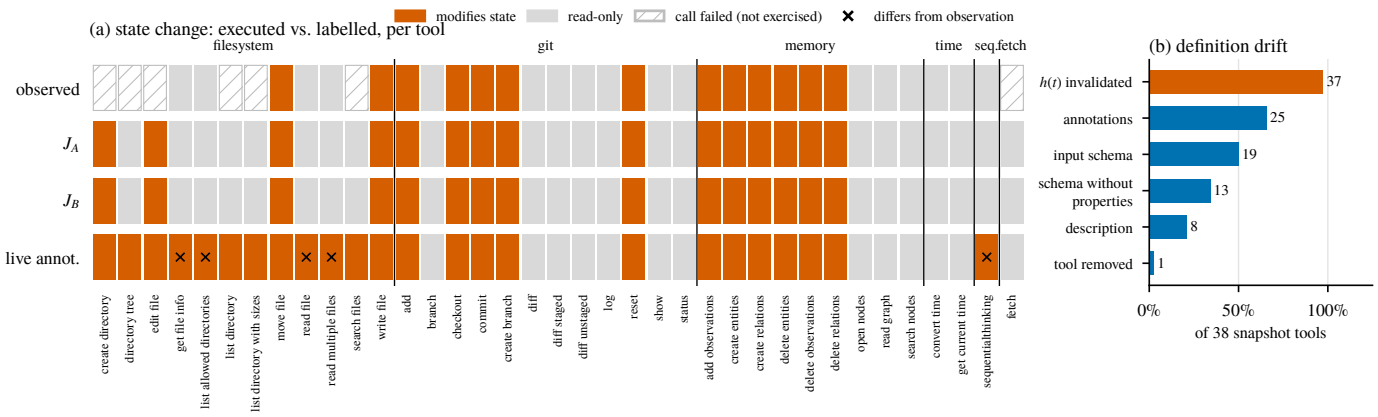


Figure 15. Executed reference servers. (a) Per registry tool: state change observed under tracing, the two labellings and the servers' current annotations (not read-only unless `readOnlyHint=true`). (b) Changes of the live definitions against the registry snapshot; every changed definition changes $h(t)$.

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